

Quiz 1 — Practice Problems

Reference: Banda-Barragán, W. E. (2026). *Computational Physics 1: Unit 1 lecture notes*. GitHub. <https://github.com/wbandabarragan/computational-physics-1/tree/main/unit-1>

How to work through it

- **The quiz will be written, and taken without a computer, a notebook or a phone.** You will answer concept questions, hand-write pseudocode and workflow diagrams.
- Reading the commands is not the same as typing them, and the ones you will be asked about are those you use every day.
- Then practise on paper: explain concepts and draw the diagrams by hand.
- Submission of these problems is not needed. This sheet is practice for Quiz 1.

Part A — What to study

- **100 — Inaugural class.** What computational physics is for.
- **101 — Google Colab and Notebooks.** Cell types, runtime, computing resources, running Linux commands and Python objects, mounting Google Drive.
- **102 — Linux and open-source software.** The filesystem tree, basic Linux commands, file permissions and `chmod`, in both the symbolic and the octal form.
- **103 — Editors and IDEs.** Available options, usage, basic functionality.
- **104 — Git, GitHub and GitLab.** Repositories, concepts and functionalities.
- **105 — Python package managers.** Available options, basic functionality, environments. How we use them, what are they good for.
- **106 — Computer science and scientific programming.** The steps of problem solving, high-level versus low-level languages, interpreter versus compiler.
- **107 — Algorithms and workflow diagrams.** The symbols and what each one means, the *what* and the *how* of a diagram, how a loop and a branch are drawn.

Part B — Workflow diagrams

Practise sketching diagrams for the problems outlined below using the symbols of Lecture 107, with a few lines of pseudocode beside it. Mark the data I/O, the operations, the decisions and the loops, and name the functions.

1. **A projectile with drag.** A ball is launched from the ground with speed v_0 at an angle θ . Draw the workflow of a program that advances its trajectory in small time steps including air drag, stops when the ball reaches the ground again, and returns the range and the time of flight.
2. **The period of a pendulum.** You measure the period T of a simple pendulum for several lengths L . Draw the workflow of a program that reads the two columns from a file, discards any measurement flagged as bad, fits T^2 against L , and returns g with its uncertainty.